

RePlate: Reinventing What Goes on the Plate

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Abstract

RePlate is a cloud-based platform that leverages Azure services to address the growing issue of food waste by connecting farmers, vendors, and consumers. Using a modern data pipeline architecture, AI-powered analytics, and a secure, scalable infrastructure, RePlate streamlines the food supply chain, reduces operational costs, and maximizes access to imperfect but edible produce. This article outlines the objectives, architecture, data engineering pipeline, and business outcomes of the RePlate initiative.

Table of Contents

1. Introduction
2. Vision and Mission
3. Objectives
4. System Architecture Overview
 - Design and Discovery Phase
5. Data Sources and Outputs
6. Data Pipeline Layers
 - Bronze Layer
 - Silver Layer
 - Gold Layer
7. Pipeline Orchestration
8. Pipeline Failure Handling
9. Business Impact and Future Directions
10. Team Contributions
11. Conclusion
12. References

Introduction

Globally, a third of all food produced is wasted. RePlate aims to address this issue by creating a digital marketplace powered by cloud and AI technologies, enabling real-time collaboration between food producers, distributors, and consumers.

Vision and Mission

Vision: To create a smart, sustainable food system using real-time insights and cloud-first design.

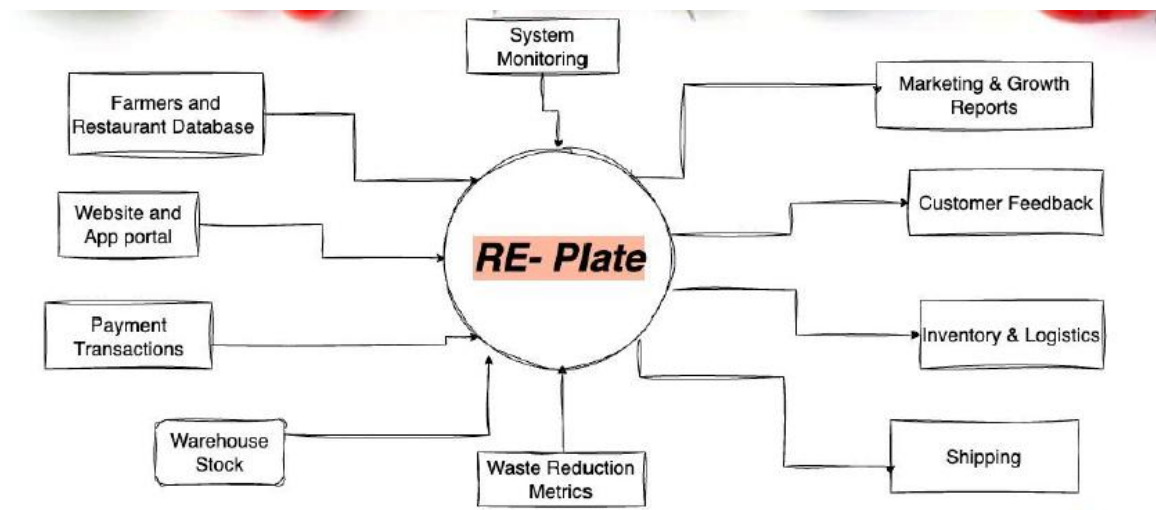
Mission: To reduce food waste by bridging supply chain gaps using data, AI, and cloud-native tools.

Objectives

- Optimize the food supply chain through real-time data tracking and AI-driven forecasting.
- Build a scalable and automated marketplace using cloud-native architecture.
- Enhance operational efficiency while maintaining strict compliance and data security.
- Enable seamless interaction between stakeholders via centralized data access and predictive insights.

System Architecture Overview

Design and Discovery Phase:



The architecture comprises batch and streaming data pipelines layered into Bronze, Silver, and Gold tiers, built on Azure services. Key components include Azure Data Factory, Event Hub, Stream Analytics, Azure Data Lake Gen2, Delta Lake, Azure Synapse Analytics, Cosmos DB, and Power BI.

Data Sources and Outputs

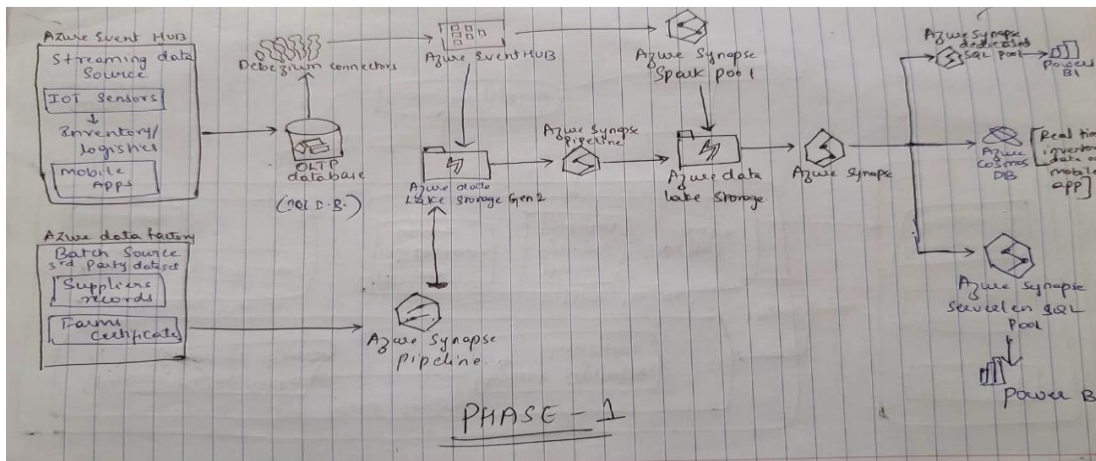
Inputs:

- Farmers and Restaurant Database
- Website & Mobile App
- Payment Transactions
- Warehouse Inventory

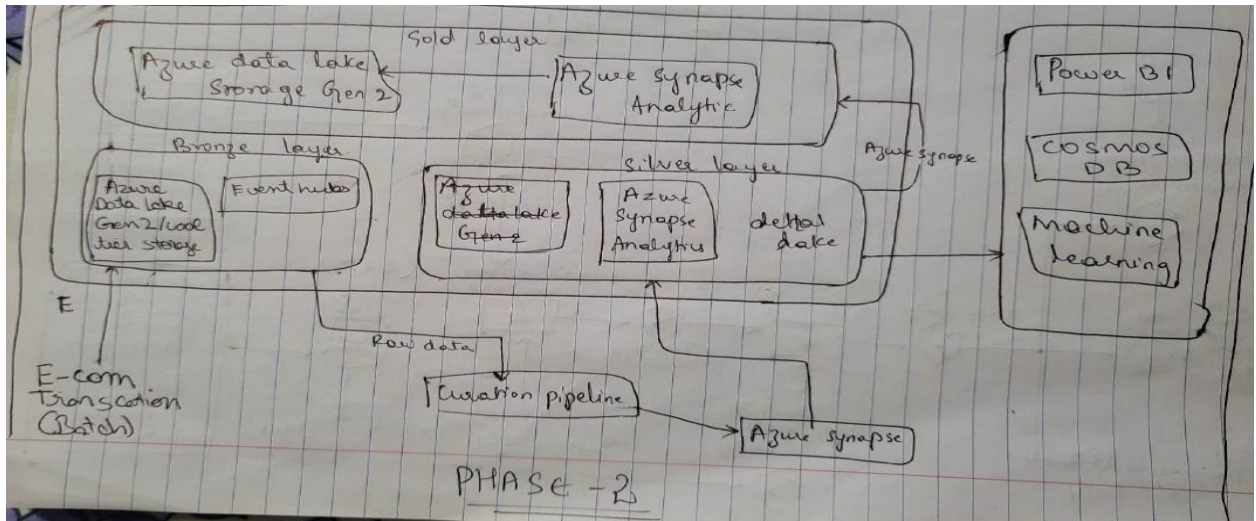
Outputs:

- | Department | Delivery Method | Key Deliverables |
- | Inventory & Logistics | Power BI, Cosmos DB, ML | Real-time tracking, demand forecasts |
- | Shipping | Power BI, ML | SLA tracking, ETAs |
- | Marketing | Power BI, ML | ROI insights, churn models |
- | Customer Feedback | NLP, Power BI | Sentiment analysis, issue trends |

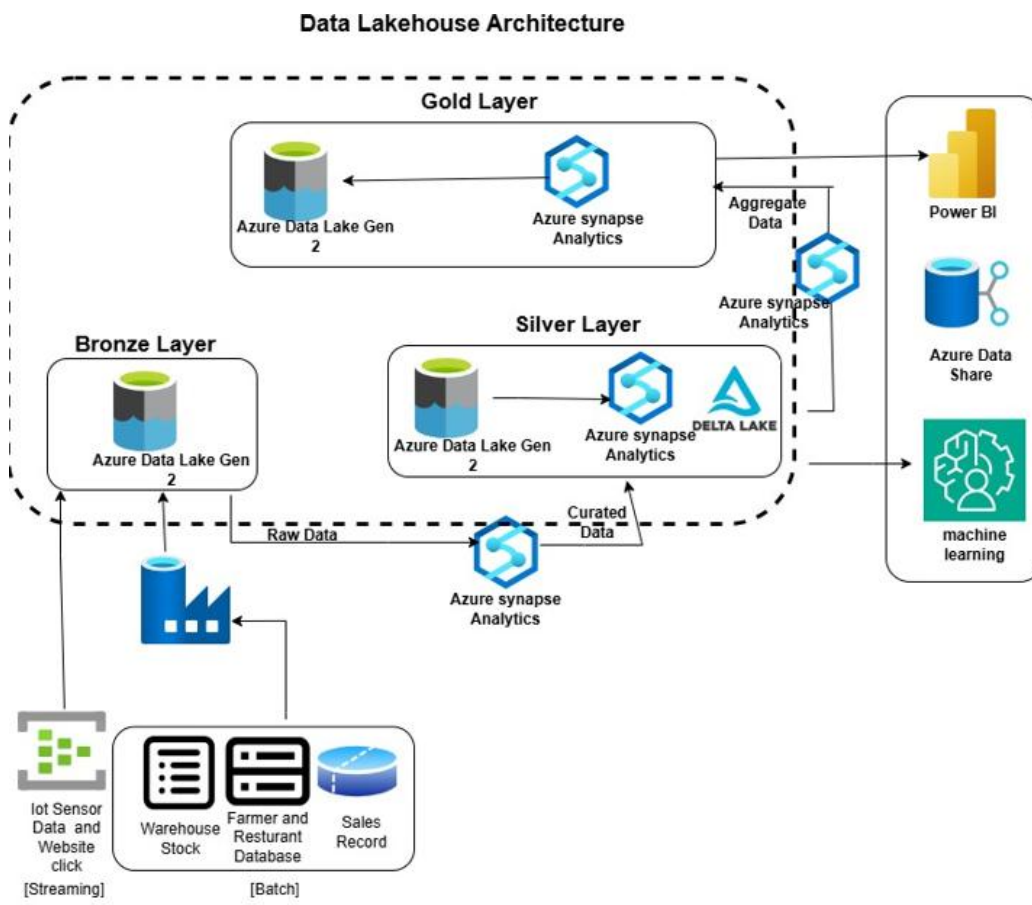
Phase 1:



Phase 2:



Final Phase 3:



Architecture Breakdown

1. Data Sources (Ingestion):

IoT Sensor Data and Website Clicks (Streaming):

Real-time data streams, such as sensor inputs and user interactions.

Real-time, direct insight into supply chain activity.

Warehouse Stock, Farmer and Restaurant Database, Sales Record (Batch):

Historical and transactional data collected periodically.

Purpose: These represent various sources feeding data into the system.

2. Bronze Layer (Raw Data Storage)

Azure Data Lake Gen2:

Stores raw data as it arrives (structured, semi-structured, unstructured).

Acts as the first landing zone.

Cost-effective, scalable object storage.

Azure Synapse:

Orchestrates data movement and transformation pipelines.

Purpose: Store raw data without transformation for future reprocessing if needed.

3. Silver Layer (Curated Data)

Azure Data Lake Gen2:

Stores cleaned, enriched, and transformed data.

Azure Synapse Analytics:

Performs transformation, joins, and business logic.

Delta Lake:

Provides ACID transactions, time travel, and schema enforcement, schema on read and schema on write.

Ensures reliability and consistency in the curated data.

Purpose: Prepare data for analytics with cleaning and joining processes.

4. Gold Layer (Aggregated/Business-Ready Data)

Azure Data Lake Gen2:

Stores aggregated or summarized data (for reporting and dashboards).

Azure Synapse Analytics:

Handles complex aggregations and data modeling.

Purpose: Final layer to serve highly optimized and summarized data for reporting.

5. Consumption Layer (Insights & Applications)

Power BI:

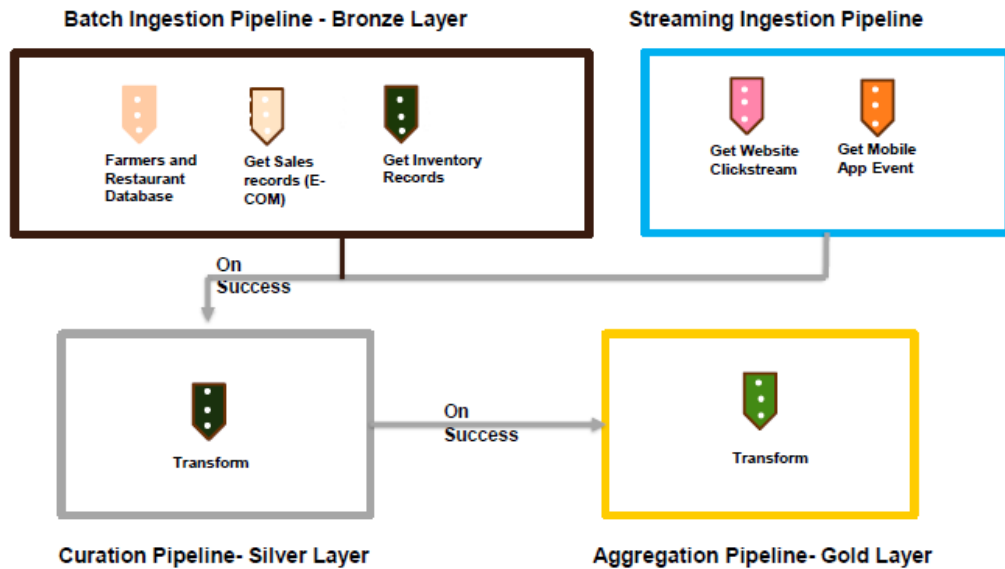
Visualizes business insights using dashboards and reports.

Machine Learning:

Uses curated and aggregated data for model training and inference.

Purpose: Deliver business value through dashboards, apps, or intelligent models.

Data Pipeline Layers



Bronze Layer: Raw data ingestion using Azure Data Factory (batch) and Event Hub + Stream Analytics (real-time). Stores raw data in Azure Data Lake Gen2 for auditability.

Silver Layer: Cleaned and curated data using Delta Lake. Steps include deduplication, schema normalization, enrichment. Stored in /silver/ layer.

Gold Layer: Aggregated business insights using Synapse and Power BI. Includes KPIs, trend analysis, customer segments.

Pipeline Orchestration

A Master Pipeline triggers sub-pipelines in a sequence (Bronze → Silver → Gold). Scheduled via Azure Data Factory and Event Hub. Ensures upstream processes complete before downstream execution.

Pipeline Failure Handling

- Common Failures: Connectivity, corrupted files, misconfiguration
- Strategy: Auto-retry, exponential backoff, logging, alerting, rollback, manual intervention if needed

Business Impact and Future Directions

- Current: Farm optimization, compliance, real-time dashboards
- Future: Integrate drone/satellite data, train ML for pricing & demand, provide APIs via Cosmos DB

Conclusion

RePlate presents a robust cloud architecture that supports end-to-end automation from ingestion to insights. Through Azure-native services and a layered data approach, it demonstrates how cloud technology can be harnessed to build a scalable, sustainable solution to food waste.

References

- Microsoft Azure Documentation
- Power BI Official Guide
- Delta Lake Guide
- Event Hub, Stream Analytics Docs